Pyrothalasa

# The Hydrogen Garden

**Website** https://omarshobaki-png.github.io/Pyrothalasa/

**Video** https://drive.google.com/file/d/1dLfS1ZW1qGngDrTg4QJEw0VGyg4zmLMQ/view

A food web under ice, powered by hydrogen

## 1. Team and selected challenge

**Team Shay Rafe3** · Challenge A, Life Without Sunlight

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## 2. Problem and intended users

Food web lessons often start with sunlight. How could a food web survive in a dark ocean under ice? We designed a classroom activity for students ages 14 to 18 and their science teachers. Cassini's hydrogen finding on Enceladus connects that question to a real ocean world. [2]

**Learning goals.** Trace energy from rock to Shadowfin, explain how one habitat condition shapes survival, and predict what happens if hydrogen stops flowing.

## 3. How the dive, video and report work together

Students scroll 50 km down our fictional moon, Pyrothalasa. They tap labels at the Hydrogen Garden and can switch off the vent. Earth's Lost City inspired the white chimneys. The video introduces the food web and this report explains the evidence. [1]

View the source code

| Habitat condition | What it means for life in the Hydrogen Garden |
|---|---|
| **Depth and pressure** | Under 20 km of ice and 30 km of ocean. Cells must work at about 56 bar, roughly 55 times the air pressure at sea level. [3] |
| **Temperature** | In our model, the open ocean is near freezing and water around the chimneys reaches about 8 °C. We place methane-makers in warmer pores (tiny holes). Lost City's vent fluid reaches 90 °C. [1] |
| **Salt and chemistry** | Cells must balance salt and water. We assume salty ocean water brings enough dissolved carbon dioxide ($CO_2$) into the pores, where it mixes with hydrogen-rich, alkaline vent water. |
| **Darkness and oxygen** | Ice blocks sunlight. We assume no dissolved oxygen reaches the garden. Microbes use chemical energy to grow in the dark. [11] |
| **Energy source** | Water reacts with iron-rich rock and releases hydrogen ($H_2$). We assume warm fluid keeps bringing $H_2$ into the same pores as the $CO_2$. [1] [2] |

**Pressure estimate.** Using Enceladus-like gravity (0.113 m/s²), the weight of the ice and water gives about 56 bar. We assume densities of 920 kg/m³ for ice and 1,030 kg/m³ for salty water. [3]

**4. Supporting science**

# Why hydrogen?

Water reacts with seafloor rock and releases hydrogen. This is **serpentinization**. In warm chimney pores, microbes use hydrogen and dissolved $CO_2$ to gain energy and grow. Using chemical energy to build living material is **chemosynthesis**. [1] [2] [6] [11]

**$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$**
Carbon dioxide + hydrogen → methane + water. Methane is a waste product.

**Why the supply could last.** The planet's changing pull can warm the moon and keep water moving through rock. A model of Enceladus shows how that could work. Earth's Lost City vents have been active for at least 30,000 years. We assume our vent keeps reaching fresh rock. [4] [5]

## Four choices and the evidence behind them

| Our design choice | Real-world evidence | What we assume |
|---|---|---|
| **Hydrogen from rock and water** | Lost City vents release hydrogen. Cassini measured it in Enceladus's plume. [1] [2] | Water keeps circulating through rock and supplies enough hydrogen. |
| **Methane-makers in warm pores** | An Earth microbe made methane at 50 bar in an Enceladus-inspired lab test. [6] | Warm pores contain enough $H_2$ and dissolved $CO_2$ for growth. |
| **A salty ocean under ice** | Salts in ice grains and Enceladus's wobble point to an ocean. Silica grains point to hot water reacting with rock. [7] [14] [15] | One vent field in our ocean has the mix of conditions shown on page 1. |
| **An oxygen-free, microscopic food web** | A study reported tiny animals called loriciferans living without oxygen. Their ability to complete a life cycle there is debated. [8] [9] | Our grazers and Shadowfin live and feed without oxygen. |

## Why we chose hydrogen

Could the option work in darkness, without oxygen, using chemicals we can justify? Hydrogen and $CO_2$ fit our model.

| Option | Why it appealed | Why we ruled it out |
|---|---|---|
| **Plants or algae** | Students know these producers. | Photosynthesis needs light. [11] |
| **Tubeworm vents** | A real food web in the dark. | The tubeworms' bacteria use oxygen to react with sulfide. [11] |
| **Methane eaters** | Our microbes release methane. | Some Earth seep microbes use sulfate. We do not know if the garden can supply it. [12] |

**5. Energy flow**

# From rock to Shadowfin

| Energy Source | Level 1 | Level 2 | Level 3 |
|---|---|---|---|
| **Vent hydrogen** | **Methane-makers** | **Grazers** | **Shadowfin** |
| Water reacts with rock and releases $H_2$. | Microbes use $H_2$ and $CO_2$ to grow. | Single cells eat microbes in cooler water. | The spiny predator eats grazers. It is under 1 mm long. |
| Chemical energy | 1,000 coins | ≈100 coins | ≈10 coins |

**Reading the numbers.** The dive starts with 1,000 energy coins in producer growth. If about 10% passes on at each step, about 10 coins reach Shadowfin. The coins illustrate an Earth average. The garden's actual transfer is unknown. Organisms use energy to live and release heat. Uneaten food, dead cells and waste can feed decomposer bacteria. [10]

**If hydrogen stops flowing,** methane-makers make less new food. Grazers then have less to eat, followed by Shadowfin. The food web could shrink or collapse.

## 6. Feasibility, limits and validation

A real ocean world could have conditions that support life. Whether any life exists there remains an open question.

**Carbon supply.** Some Lost City vent fluids contain little dissolved inorganic carbon. We assume enough $CO_2$ from the ocean reaches the same pores as $H_2$. [13]

**Survival and feeding.** We do not yet know whether grazers can grow on the vent microbes in salty, oxygen-free water or whether a predator like Shadowfin could live there. [8] [9]

**Energy budget.** We have no measurement of the garden's hydrogen supply or how much energy reaches the predator. [10]

**First tester.** One 18-year-old tried the site and watched the video. They found both very entertaining. This was feedback about enjoyment.

**Next classroom check.** Ask a science teacher to review the dive. Have five students ages 14 to 18 try its short quiz before and after. After the dive, aim for four of five to trace the energy path, explain one survival condition and predict what happens if hydrogen stops. Revise labels that confuse several students.

**Lab check.** A microbiology lab could grow Earth methane-makers in matched warm, salty, oxygen-free chambers at about 56 bar, with and without $H_2$. More methane and more cells over time with $H_2$ than without it would support the first living level. A chamber with no cells would check whether methane forms without microbes. A later feeding test would check whether grazers survive and reproduce on those microbes. [6]

**What to look for in a real vent.** Finding $H_2$ and dissolved $CO_2$ together in warm, oxygen-free water would make the habitat more plausible.

**7. Contributions and sources**

# Teamwork and references

**Omar Shoubaki** · Led the science research on the habitat and hydrogen pathway.

**Ahmad Sulieman** · Led evidence checking and validation planning, and supported science research.

**Omar Takhman** · Co-led the science work and contributed to the food web and video script.

**Alameen Abufkheida** · Led ecology discussions and helped write the report.

**Mahdi Alsheikh** · Edited the team video.

**Mohammad Humidan** · Programmed the interactive dive website.

**How the design changed.** An early science mock-up considered oxygen in cooler water. We settled on a microscopic food web that runs without oxygen throughout the garden, then added the evidence and limits for that choice.

## AI use and attribution

**ChatGPT (Codex)** helped with writing and formatting the report. **Claude** reviewed the report against the rubric. **Claude Code** assisted with programming and **ElevenLabs** with narration audio. Mahdi edited the video in **Adobe After Effects**.

One tester gave feedback on the site and video.

**Credits.** The team created Pyrothalasa, the Hydrogen Garden and its organisms. The diagram follows the team's model. Research and agency sources are credited below.

## References


**[1] Kelley et al. (2005).** A serpentinite-hosted ecosystem: The Lost City Hydrothermal Field. *Science.*

**[2] Waite et al. (2017).** Cassini finds molecular hydrogen in the Enceladus plume: Evidence for hydrothermal processes. *Science.*

**[3] NASA/JPL.** Planetary Satellite Physical Parameters. *Enceladus mass parameter and mean radius.*

**[4] Choblet et al. (2017).** Powering prolonged hydrothermal activity inside Enceladus. *Nature Astronomy.*

**[5] Früh-Green et al. (2003).** 30,000 years of hydrothermal activity at the Lost City vent field. *Science.*

**[6] Taubner et al. (2018).** Biological methane production under putative Enceladus-like conditions. *Nature Communications.*

**[7] Postberg et al. (2009).** Sodium salts in E-ring ice grains from an ocean below the surface of Enceladus. *Nature.*

**[8] Danovaro et al. (2010).** The first metazoa living in permanently anoxic conditions. *BMC Biology.*

**[9] Bernhard et al. (2015).** Metazoans of redoxcline sediments in Mediterranean deep-sea hypersaline anoxic basins. *BMC Biology.*

**[10] Pauly and Christensen (1995).** Primary production required to sustain global fisheries. *Nature.*

**[11] NOAA Ocean Exploration.** Chemosynthesis Fact Sheet. *Deep Ocean Education Project.*

**[12] Boetius et al. (2000).** A marine microbial consortium apparently mediating anaerobic oxidation of methane. *Nature.*

**[13] Brazelton et al. (2006).** Methane- and sulfur-metabolizing microbial communities dominate the Lost City Hydrothermal Field ecosystem. *Applied and Environmental Microbiology.*

**[14] Thomas et al. (2016).** Enceladus's measured physical libration requires a global subsurface ocean. *Icarus.*

**[15] Hsu et al. (2015).** Ongoing hydrothermal activities within Enceladus. *Nature.*